ECS 152A Discussion

Week 4

Quiz 1 Review

Two browser tabs on the same host connect to the same web server IP address on port 80. How are these two connections distinguished at the client side?

A. By using different client source ports

B. By using different destination ports on the server

C. By using different server IP addresses automatically

D. By using different HTTP versions only

Doubt 1: Why can two browser tabs create different TCP connections?

A TCP connection is identified by a 4-tuple:

- source IP address
- source port number
- destination IP address
- destination port number

Same destination does not automatically mean same connection.

Worked example: two browser tabs to one web server

What changes, and what stays the same?

Tab A connection

Source IP	10.0.0.5
Source port	53001
Destination IP	128.120.x.x
Destination port	80

Tab B connection

Source IP	10.0.0.5
Source port	53002
Destination IP	128.120.x.x
Destination port	80

Web server port 80

What is different?

Only the source port changed: 53001 vs 53002. That is enough to make these two distinct TCP connections.

What stays the same?

The same client machine can talk to the same server IP and the same server port 80 at the same time.

Quick check: same server, same port 80 — same connection?

Client opens two tabs to the same website.

Both use the same destination IP and destination port 80.

Can they still be two different TCP connections?

Yes — the OS can assign different source ports.

Wrong shortcut:
Same destination = same connection

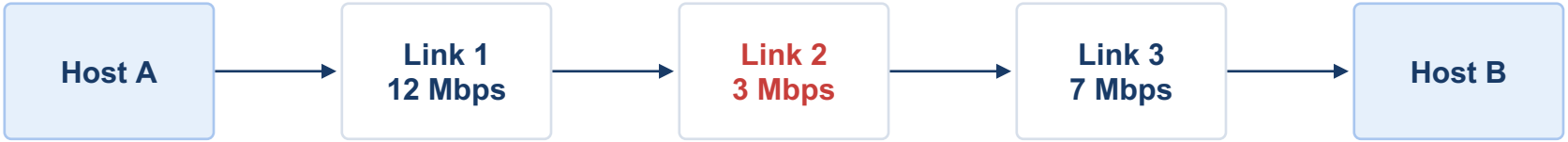
Correct rule:
All 4 fields matter

A path contains three links with capacities 12 Mbps, 7 Mbps, and 3 Mbps. What is the end-to-end throughput?

A. 12 Mbps
B. 7 Mbps
C. 3 Mbps
D. 22 Mbps

Doubt 2: why does the bottleneck determine throughput?

Key idea: data must pass every link on the path, so the slowest one limits the rate.



Wrong: 12 Mbps + 3 Mbps + 7 Mbps

But the file is not duplicated across paths. It must go through all three links in sequence.

Correct rule

End-to-end throughput = minimum link rate along the path = min(R1, R2, R3)

For this example

min(12 Mbps, 3 Mbps, 7 Mbps) = 3 Mbps

Worked example: compute transfer time after finding throughput

Step 1: find throughput

Path rates:

- R1 = 12 Mbps
- R2 = 3 Mbps
- R3 = 7 Mbps

Throughput = min(R1, R2, R3) = 3 Mbps

Step 2: convert file size

File size = 3 × 10^6 bytes
= 24 × 10^6 bits

Why convert?
Because throughput is in bits per second.

Step 3: divide

Transfer time ≈ file size / throughput

= (24 × 10^6 bits) / (3 × 10^6 bits/s)

= 8 s

Memory line

You can only send as fast as the slowest link can forward the data.

Network Performance Examples

Problem 1: How long does it take a packet of length 1,000 bytes to **propagate** over a link of distance 2,500 km, propagation speed 2.5×10^8 m/s, and transmission rate 2 Mbps? More generally, how long does it take a packet of length L to propagate over a link of distance d, propagation speed s, and transmission rate R bps? Does this delay depend on packet length? Does this delay depend on transmission rate?

What people often mix up

Propagation delay vs. transmission delay

Key reminder

Propagation = travel time on the link; Transmission = time to push bits out

Problem 1 analysis

Separate propagation delay from transmission delay

Step 1: Propagation delay

Formula:

$d_prop = d / s$

Plug in values:

$d = 2.5 \times 10^6$ m

$s = 2.5 \times 10^8$ m/s

Result:

$d_prop = 0.01$ s = 10 ms

Step 2: Transmission delay

Formula:

$d_trans = L / R$

L = 1000 bytes = 8000 bits

$R = 2 \times 10^6$ bps

Result:

$d_trans = 0.004$ s = 4 ms

Key conclusions

- Propagation delay depends on distance and propagation speed.
- It does NOT depend on packet length.
- It does NOT depend on transmission rate.
- For this question, propagation delay = 10 ms.

Memory line:

transmission is pushing bits onto the link; propagation is the signal traveling along the link.

Problem 2: Assume n packets of length **L** come into a router at the same time. The transmission rate of the router is **R.** Ignore the processing delay and calculate the average queuing delay.

Hint

If all packets arrive together, packet 1 does not wait.
Packet 2 waits for 1 transmission time, packet 3 waits for 2 transmission times, and so on.

Think in units of one transmission time: L/R.

Problem 2 analysis

Average queuing delay when n packets arrive together

Each packet's waiting time

Packet 1: 0
Packet 2: L/R
Packet 3: 2L/R
⋮
Packet n: (n−1)L/R

Total waiting time

$0 + L/R + 2L/R + \cdots + (n-1)L/R$

$= (L/R) \cdot [0 + 1 + 2 + \cdots + (n-1)]$

$= (L/R) \cdot n(n-1)/2$

Average queuing delay

Divide total by n packets:

Average = $[(L/R) \cdot n(n-1)/2] / n$

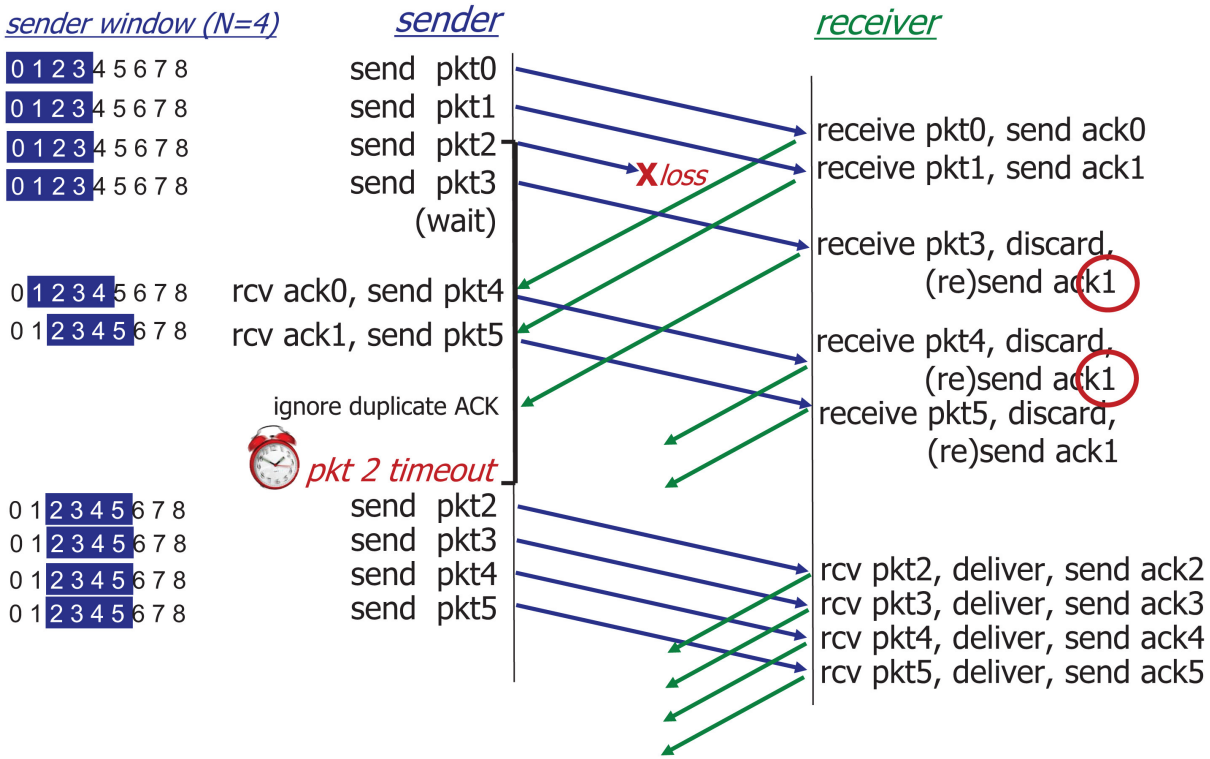
$= (n-1)L / (2R)$

Final answer:
average queuing delay = $(n-1)L/(2R)$

Memory line:
the average of 0, 1, 2, …, n−1 is (n−1)/2, so the average delay is (n−1)/2 transmission times.

GBN vs. SR

Go-Back-N in Action

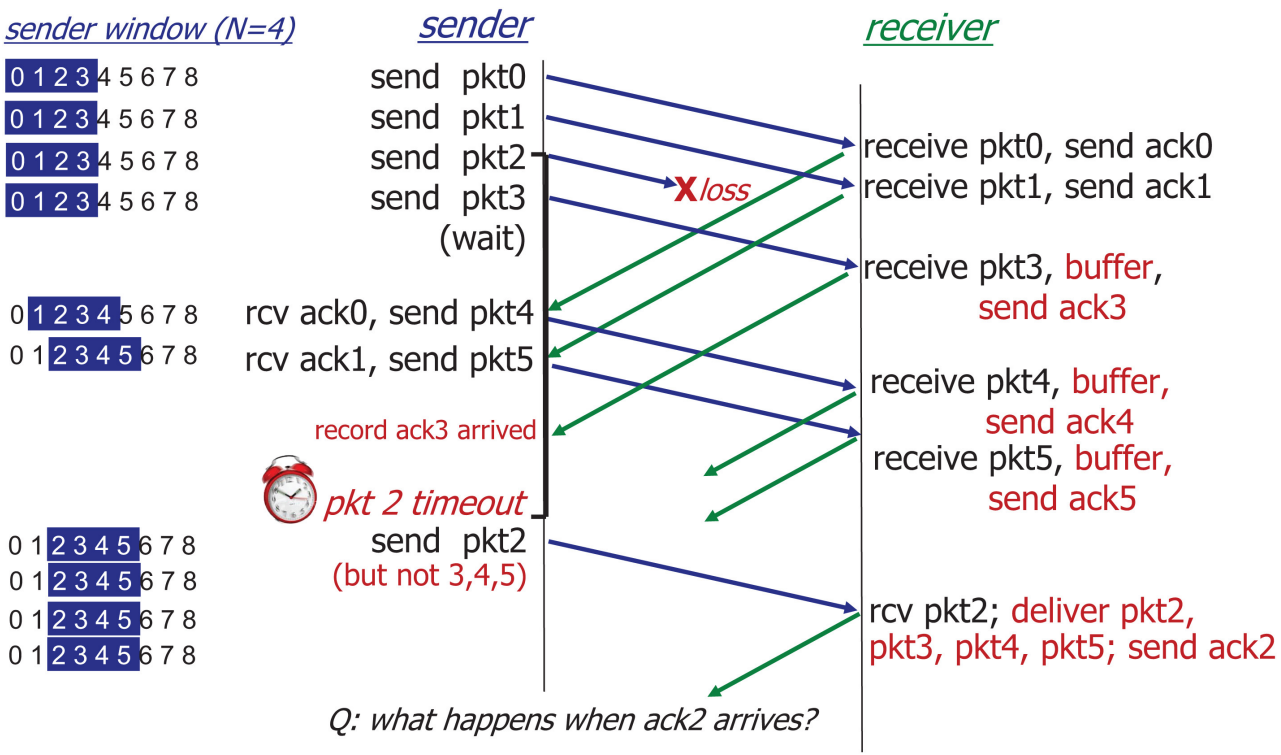


Go-Back-N in Action

Key points:

1. Receiver has no buffers for out-of-order.
2. Single lost packet causes retransmission of potentially many packets.
3. **Sender only needs one timer --** for the oldest unacknowledged packet.
4. **Cumulative ACKs** mean that if ACK(5) arrives, packets 0 through 5 are all confirmed.

Selective Repeat in Action



Selective Repeat in Action

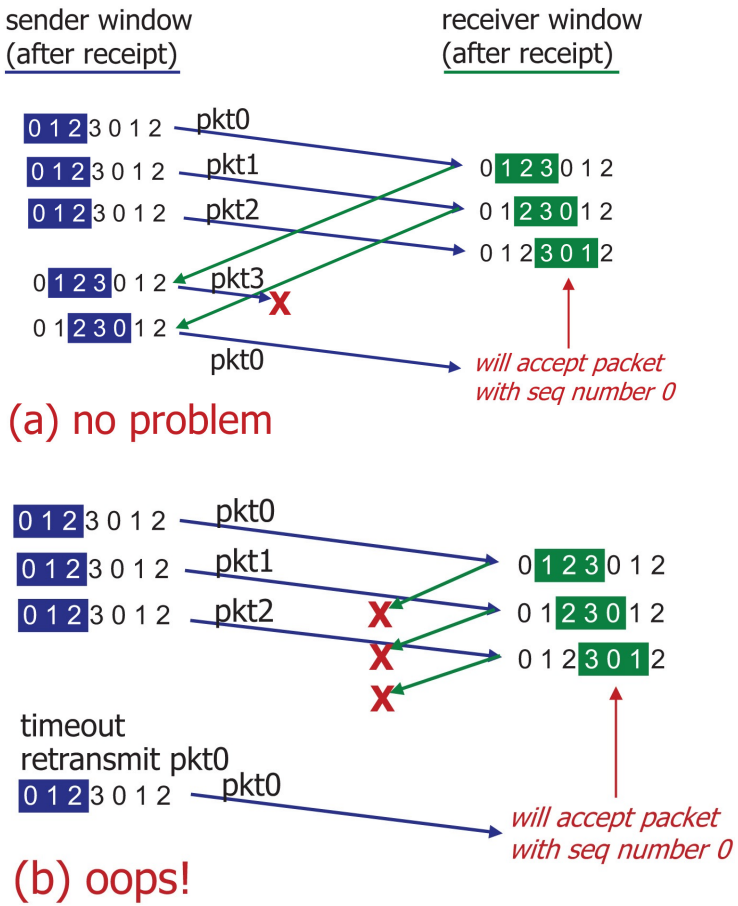
Key points:

1. Receiver maintains a buffer -- **more memory needed.**
2. Only the lost packet is retransmitted -- much more efficient.
3. **Each packet needs its own timer** -- more complexity at the sender.
4. **Individual ACKs** instead of cumulative -- the sender knows exactly which packets were received.

Selective Repeat: a Dilemma! (1 of 2)

example:

- seq #s: 0, 1, 2, 3 (base 4 counting)
- window size=3

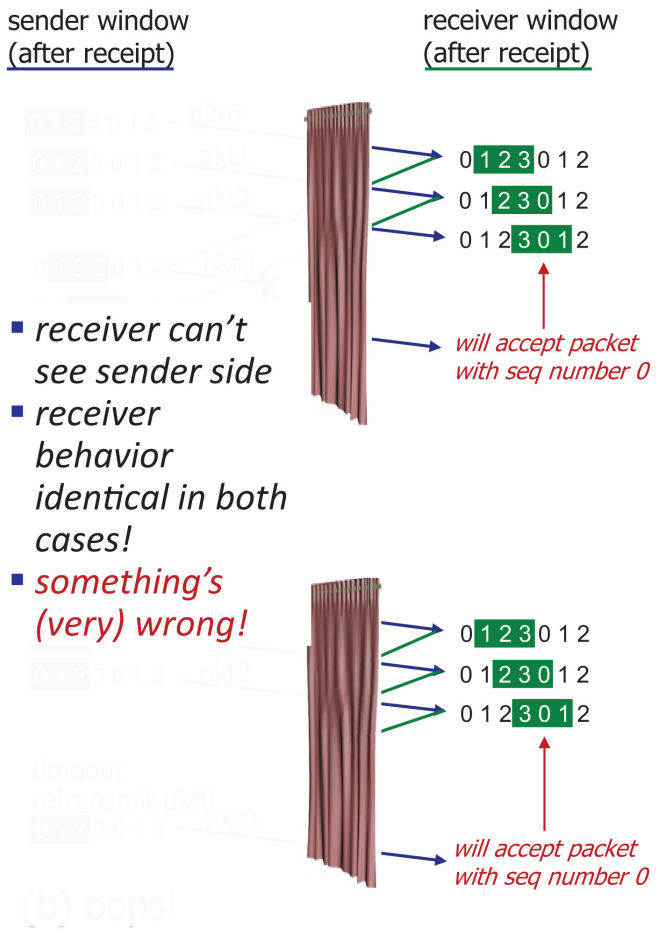


Selective Repeat: a Dilemma! (2 of 2)

example:

- seq #s: 0, 1, 2, 3 (base 4 counting)
- window size=3

Q: what relationship is needed between sequence # size and window size to avoid problem in scenario (b)?



SR Dilemma: Answer

The problem:

Sender's window and receiver's window overlap in sequence number space.

Receiver cannot distinguish old retransmissions from new packets.

The rule:

Window size W must satisfy: $\mathbf{W \leq k / 2}$

where k = number of distinct sequence numbers.

Check our example: k=4, W=3. 2×3=6 ≤ 4? **NO — rule violated!**

Window Size vs. Sequence Number Space

Let k = number of distinct sequence numbers (e.g., k = 2^n for n-bit seq #).

Go-Back-N: **$W \leq k - 1$**

Receiver only accepts in-order → less room for confusion.

Selective Repeat: $W \leq k / 2$

Receiver buffers out-of-order → stricter constraint to prevent ambiguity.

Example (3-bit seq #, k = 8): GBN: $W \leq 7$. SR: $W \leq 4$.

Takeaway: SR is stricter — buffering creates more overlap potential.